

Create Highly Available Architecture by Distributing Incoming Traffic among Healthy Service Instances in Cloud Services or Virtual Machines in a Load Balanced Set with the Help of Command-Line Interface

Course-end Project 2

Description

Project Agenda

To create high available architecture by distributing incoming traffic among healthy service instances in cloud services or virtual machines in a load-balanced set with the help of a command-line interface

Description:

The Rand Enterprises Corporation wants to deploy a web application in a highly available environment so that only the healthy instances will be serving the traffic so end users will not be facing any downtime. They have decided to work on an Azure public load balancer to implement the functionality.

The operations team at Rand decides to define the entire architecture using the load balancer and its backend pool, once that's in place they intend to create the frontend IP and health probe along with virtual machines housing their application.

Rand Enterprises works extensively on delivering highly available web applications for their users in a secure way by avoiding directly exposing the virtual machines hosting the applications to the public internet. The communication from the application in the VM to the end-user must take place via the Load Balancer.

The expectation of the operation team is to create a reusable method that can be used for automation if in the future we need to deploy the same kind of infrastructure. So, rather than deploying resources in the Azure portal, they should leverage the command-line interface to deploy the resources so that in the future these commands can be used

As a security measure, you need to ensure that only the health instances of the virtual machine will be serving the traffic.

Tools required:

Azure account with administrator access

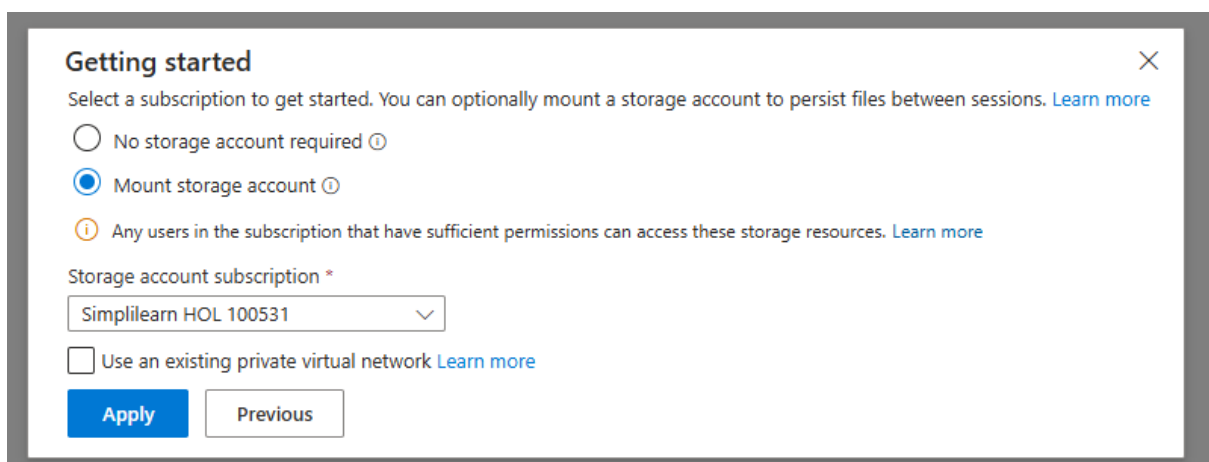
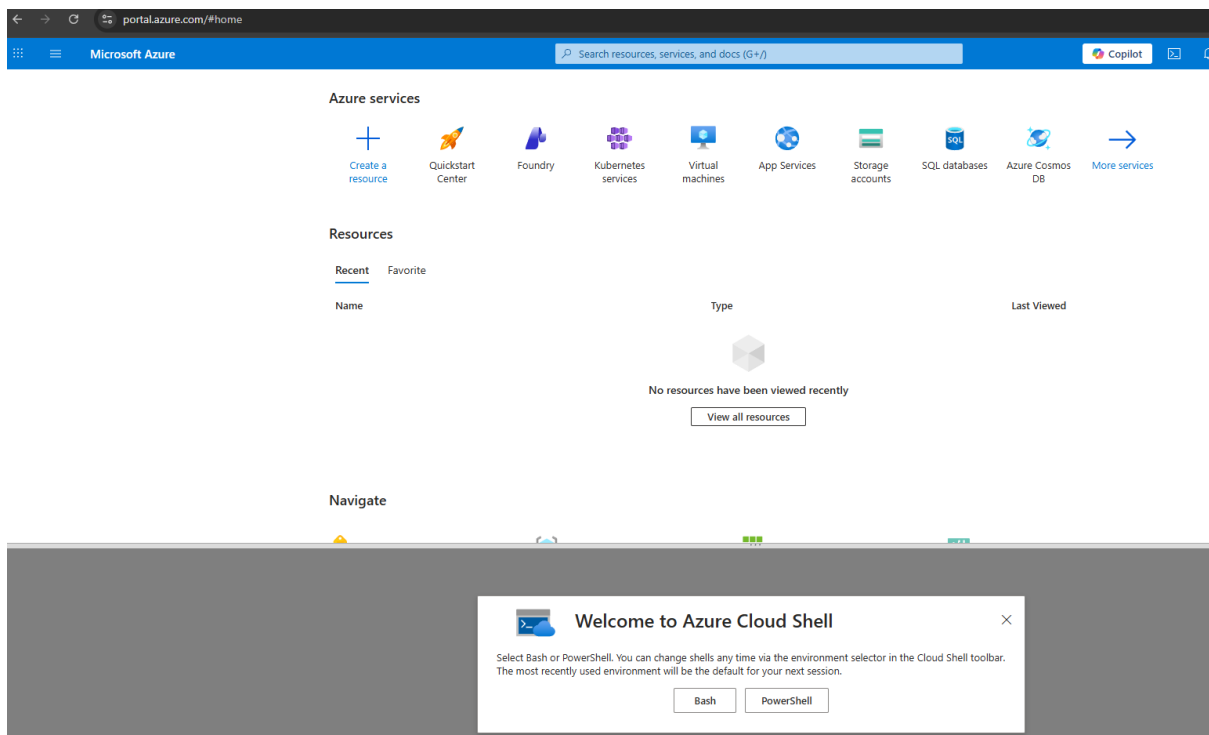
Prerequisites:

None

Expected Deliverables:

- Identify Virtual machines and Networking
- Configure the load balancer
- Extend the load balancer with backend pool and frontend IP
- Define the Health probe

Identify Virtual machines and Networking



Mount storage account



Azure Cloud Shell requires a storage account with Azure file share to persist files. Select an option below to mount a storage account. [Learn more](#)

- ☐ Select existing storage account
- ☒ We will create a storage account for you ⓘ
- ☐ I want to create a storage account

Next

Previous

Deployment is in progress

Subscription: Simplilearn HOL 100531
Resource group: baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourcegroups/cloud-shell-storage-centralindia
Storage account: csg10032005ebc744b8
File share: cs-odl-user-2240269-simplilearnhol100531-onmic-10032005ebc744b8
Region: Central India

Switch to PowerShell Restart Manage files ▾

```
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

odl_user [ ~ ]$
```

Create a resource group

```
az group create --name CreatePubLBQS-rg --location eastus
```

```
odl_user [ ~ ]$ az group create --name CreatePubLBQS-rg --location eastus
{
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg",
  "location": "eastus",
  "managedBy": null,
  "name": "CreatePubLBQS-rg",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
odl_user [ ~ ]$
```

Create a virtual network

```
az network vnet create --resource-group CreatePubLBQS-rg --location eastus --name myVNet --address-prefixes 10.1.0.0/16 --subnet-name myBackendSubnet --subnet-prefixes 10.1.0.0/24
```

```
odl_user [ ~ ]$ az network vnet create --resource-group CreatePubLBQS-rg --location eastus --name myVNet --address-prefixes 10.1.0.0/16 --subnet-name myBackendSubnet --subnet-prefixes 10.1.0.0/24
{
  "newVNet": {
    "addressSpace": {
      "addressPrefixes": [
        "10.1.0.0/16"
      ]
    },
    "enableDdosProtection": false,
    "etag": "W/\"3a5956b9-05b7-45e2-a041-d0ff4105b25e\"",
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/virtualNetworks/myVNet",
    "location": "eastus",
    "name": "myVNet",
    "privateEndpointVNetPolicies": "Disabled",
    "provisioningState": "Succeeded",
    "resourceGroup": "CreatePubLBQS-rg",
    "resourceGuid": "ab3c7913-3c83-4671-8b3b-76dc4a05a4dc",
    "subnets": [
      {
        "addressPrefix": "10.1.0.0/24",
        "delegations": [],
        "etag": "W/\"3a5956b9-05b7-45e2-a041-d0ff4105b25e\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/virtualNetworks/myVNet/subnets/myBackendSubnet",
        "name": "myBackendSubnet",
        "privateEndpointNetworkPolicies": "Disabled",
        "privateLinkServiceNetworkPolicies": "Enabled",
        "provisioningState": "Succeeded",
        "resourceGroup": "CreatePubLBQS-rg",
        "type": "Microsoft.Network/virtualNetworks/subnets"
      }
    ]
  },
  "type": "Microsoft.Network/virtualNetworks",
  "virtualNetworkPeerings": []
}
```

Create a public IP address

```
az network public-ip create --resource-group CreatePubLBQS-rg --name myPublicIP --sku Standard --zone 1
```

```
odl_user [ ~ ]$ az network public-ip create --resource-group CreatePubLBQS-rg --name myPublicIP --sku Standard --zone
1
{
  "publicIp": {
    "ddosSettings": {
      "protectionMode": "VirtualNetworkInherited"
    },
    "etag": "W/\"e60e906e-6e62-4209-ab03-21430c91af2d\"",
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Ne
twork/publicIPAddresses/myPublicIP",
    "idleTimeoutInMinutes": 4,
    "ipAddress": "52.249.219.74",
    "ipTags": [],
    "location": "eastus",
    "name": "myPublicIP",
    "provisioningState": "Succeeded",
    "publicIPAddressVersion": "IPv4",
    "publicIPAllocationMethod": "Static",
    "resourceGroup": "CreatePubLBQS-rg",
    "resourceGuid": "be03005a-af5c-40ae-b05a-64494a3300ee",
    "sku": {
      "name": "Standard",
      "tier": "Regional"
    },
    "type": "Microsoft.Network/publicIPAddresses",
    "zones": [
      "1"
    ]
  }
}
odl_user [ ~ ]$
```

Configure the load balancer

Create the load balancer resource

```
az network lb create --resource-group CreatePubLBQS-rg --name myLoadBalancer --sku Standard --public-ip-address myPublicIP --frontend-ip-name myFrontEnd --public-ip-zone 1 --backend-pool-name myBackEndPool
```

```
odl_user [ ~ ]$ az network lb create --resource-group CreatePubLBQS-rg --name myLoadBalancer --sku Standard --public-ip-address myPublicIP --frontend-ip-name myFrontEnd --public-ip-zone 1 --backend-pool-name myBackEndPool
{
  "loadBalancer": {
    "backendAddressPools": [
      {
        "etag": "W/\"dbbc0fc6-68fc-431c-8ad9-81129a01b278\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/backendAddressPools/myBackEndPool",
        "name": "myBackEndPool",
        "properties": {
          "loadBalancerBackendAddresses": [],
          "provisioningState": "Succeeded"
        },
        "resourceGroup": "CreatePubLBQS-rg",
        "type": "Microsoft.Network/loadBalancers/backendAddressPools"
      }
    ],
    "frontendIPConfigurations": [
      {
        "etag": "W/\"dbbc0fc6-68fc-431c-8ad9-81129a01b278\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/frontendIPConfigurations/myFrontEnd",
        "name": "myFrontEnd",
        "properties": {
          "privateIPAllocationMethod": "Dynamic",
          "provisioningState": "Succeeded",
          "publicIPAddress": {
            "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/publicIPAddresses/myPublicIP",
            "resourceGroup": "CreatePubLBQS-rg"
          }
        },
        "resourceGroup": "CreatePubLBQS-rg",
        "type": "Microsoft.Network/loadBalancers/frontendIPConfigurations"
      }
    ],
    "inboundNatPools": [],
    "inboundNatRules": []
  }
}
```

Create the health probe

```
az network lb probe create --resource-group CreatePubLBQS-rg --lb-name myLoadBalancer --name myHealthProbe --protocol tcp --port 80
```

```
odl_user [ ~ ]$ az network lb probe create --resource-group CreatePubLBQS-rg --lb-name myLoadBalancer --name myHealthProbe --protocol tcp --port 80
{
  "etag": "W/\"608ca329-10b9-4658-8010-8d1645188654\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/probes/myHealthProbe",
  "intervalInSeconds": 15,
  "name": "myHealthProbe",
  "numberOfProbes": 2,
  "port": 80,
  "probeThreshold": 1,
  "protocol": "Tcp",
  "provisioningState": "Succeeded",
  "resourceGroup": "CreatePubLBQS-rg",
  "type": "Microsoft.Network/loadBalancers/probes"
}
```

Create the load balancer rule

```
az network lb rule create --resource-group CreatePubLBQS-rg --lb-name myLoadBalancer --name myHTTPRule --protocol tcp --frontend-port 80 --backend-port 80 --frontend-ip-name myFrontEnd --backend-pool-name myBackEndPool --probe-name myHealthProbe --disable-outbound-snat true --idle-timeout 15 --enable-tcp-reset true
```

```
odl_user [ ~ ]$ az network lb rule create --resource-group CreatePubLBQS-rg --lb-name myLoadBalancer --name myHTTPRule --protocol tcp --frontend-port 80 --backend-port 80 --frontend-ip-name myFrontEnd --backend-pool-name myBackEndPool --probe-name myHealthProbe --disable-outbound-snat true --idle-timeout 15 --enable-tcp-reset true
{
  "backendAddressPool": {
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/backendAddressPools/myBackEndPool",
    "resourceGroup": "CreatePubLBQS-rg"
  },
  "backendAddressPools": [
    {
      "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/backendAddressPools/myBackEndPool",
      "resourceGroup": "CreatePubLBQS-rg"
    }
  ],
  "backendPort": 80,
  "disableOutboundSnat": true,
  "enableFloatingIP": false,
  "enableTcpReset": true,
  "etag": "W/\"d2752421-3611-44db-97b7-d70a26b7f85f\"",
  "frontendIPConfiguration": {
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/frontendIPConfigurations/myFrontEnd",
    "resourceGroup": "CreatePubLBQS-rg"
  },
  "frontendPort": 80,
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/loadBalancingRules/myHTTPRule",
  "idleTimeoutInMinutes": 15,
  "loadDistribution": "Default",
  "name": "myHTTPRule",
  "probe": {
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/probes/myHealthProbe",
    "resourceGroup": "CreatePubLBQS-rg"
  },
  "protocol": "Tcp",
}
```

Create a network security group

```
az network nsg create --resource-group CreatePubLBQS-rg --name myNSG
```

```

odl_user [ ~ ]$ az network nsg create --resource-group CreatePubLBQS-rg --name myNSG
{
  "NewNSG": {
    "defaultSecurityRules": [
      {
        "access": "Allow",
        "description": "Allow inbound traffic from all VMs in VNET",
        "destinationAddressPrefix": "VirtualNetwork",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "*",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"57b14ee1-b29b-4e51-b376-28cd77d83398\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/networkSecurityGroups/myNSG/defaultSecurityRules/AllowVnetInBound",
        "name": "AllowVnetInBound",
        "priority": 65000,
        "protocol": "*",
        "provisioningState": "Succeeded",
        "resourceGroup": "CreatePubLBQS-rg",
        "sourceAddressPrefix": "VirtualNetwork",
        "sourceAddressPrefixes": [],
        "sourcePortRange": "*",
        "sourcePortRanges": [],
        "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
      },
      {
        "access": "Allow",
        "description": "Allow inbound traffic from azure load balancer",
        "destinationAddressPrefix": "*",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "*",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"57b14ee1-b29b-4e51-b376-28cd77d83398\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/networkSecurityGroups/myNSG/defaultSecurityRules/AllowAzureLoadBalancerInBound",

```

Create a network security group rule

```

az network nsg rule create --resource-group CreatePubLBQS-rg --nsg-name myNSG --name
myNSGRuleHTTP --protocol '*' --direction inbound --source-address-prefix '*' --source-port-range
'*' --destination-address-prefix '*' --destination-port-range 80 --access allow --priority 200

```

```

odl_user [ ~ ]$ az network nsg rule create --resource-group CreatePubLBQS-rg --nsg-name myNSG --name myNSGRuleHTTP --
protocol '*' --direction inbound --source-address-prefix '*' --source-port-range '*' --destination-address-prefix '*'
--destination-port-range 80 --access allow --priority 200
{
  "access": "Allow",
  "destinationAddressPrefix": "*",
  "destinationAddressPrefixes": [],
  "destinationPortRange": "80",
  "destinationPortRanges": [],
  "direction": "Inbound",
  "etag": "W/\"a7e9644f-5aef-4e33-958e-e518679af14f\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/networkSecurityGroups/myNSG/securityRules/myNSGRuleHTTP",
  "name": "myNSGRuleHTTP",
  "priority": 200,
  "protocol": "*",
  "provisioningState": "Succeeded",
  "resourceGroup": "CreatePubLBQS-rg",
  "sourceAddressPrefix": "*",
  "sourceAddressPrefixes": [],
  "sourcePortRange": "*",
  "sourcePortRanges": [],
  "type": "Microsoft.Network/networkSecurityGroups/securityRules"
}
odl_user [ ~ ]$

```


Create backend servers

Create network interfaces for the virtual machines

```
array=(myNicVM1 myNicVM2)
for vmnic in "${array[@]}"
do
    az network nic create --resource-group CreatePubLBQS-rg --name $vmnic --vnet-name myVNet -
--subnet myBackEndSubnet --network-security-group myNSG
done
```

```
odl_user [ ~ ]$ array=(myNicVM1 myNicVM2)
for vmnic in "${array[@]}"
do
    az network nic create --resource-group CreatePubLBQS-rg --name $vmnic --vnet-name myVNet --subnet myBackEndSubnet
--network-security-group myNSG
done
{
  "NewNIC": {
    "auxiliaryMode": "None",
    "auxiliarySku": "None",
    "disableTcpStateTracking": false,
    "dnsSettings": {
      "appliedDnsServers": [],
      "dnsServers": [],
      "internalDomainNameSuffix": "cn2tzk2dhryuncz1o1oeubne1e.bx.internal.cloudapp.net"
    },
    "enableIPForwarding": false,
    "etag": "W/\"da12a52c-4851-4f03-897f-591b84670abc\"",
    "hostedWorkloads": [],
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Ne
twork/networkInterfaces/myNicVM1",
    "ipConfigurations": [
      {
        "etag": "W/\"da12a52c-4851-4f03-897f-591b84670abc\"",
        "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsof
t.Network/networkInterfaces/myNicVM1/ipConfigurations/ipconfig1",
        "name": "ipconfig1",
        "primary": true,
        "privateIPAddress": "10.1.0.4",
        "privateIPAddressVersion": "IPv4",
        "privateIPAllocationMethod": "Dynamic",
        "provisioningState": "Succeeded",
        "resourceGroup": "CreatePubLBQS-rg",
        "subnet": {
          "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Micros
```

Create virtual machines

VM1

```
az vm create --resource-group CreatePubLBQS-rg --name myVM1 --nics myNicVM1 --image
win2019datacenter --size Standard_B1s --storage-sku Standard_LRS --admin-username azureuser --
zone 1 --no-wait
```

```
odl_user [ ~ ]$ az vm create --resource-group CreatePubLBQS-rg --name myVM1 --nics myNicVM1 --image win2019datacenter
--size Standard_B1s --storage-sku Standard_LRS --admin-username azureuser --zone 1 --no-wait
The default value of '--size' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
Admin Password:
Confirm Admin Password:
Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, ple
ase visit https://aka.ms/TrustedLaunch.
odl_user [ ~ ]$
```

VM2

az vm create --resource-group CreatePubLBQS-rg --name myVM2 --nics myNicVM2 --image win2019datacenter --size Standard_B1s --storage-sku Standard_LRS --admin-username azureuser --zone 2 --no-wait

```
odl_user [ ~ ]$ az vm create --resource-group CreatePubLBQS-rg --name myVM2 --nics myNicVM2 --image win2019datacenter
--size Standard_B1s --storage-sku Standard_LRS --admin-username azureuser --zone 2 --no-wait
The default value of '--size' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
Admin Password:
Confirm Admin Password:
Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, please visit https://aka.ms/TrustedLaunch.
odl_user [ ~ ]$
```

Add virtual machines to load balancer backend pool

```
array=(myNicVM1 myNicVM2)
for vmnic in "${array[@]}"
do
    az network nic ip-config address-pool add --address-pool myBackendPool --ip-config-name ipconfig1 --nic-name $vmnic --resource-group CreatePubLBQS-rg --lb-name myLoadBalancer
done
```

```
odl_user [ ~ ]$ array=(myNicVM1 myNicVM2)
for vmnic in "${array[@]}"
do
    az network nic ip-config address-pool add --address-pool myBackendPool --ip-config-name ipconfig1 --nic-name $vmnic
--resource-group CreatePubLBQS-rg --lb-name myLoadBalancer
done
{
  "etag": "W/\"a2cf365f-0cd5-4a35-904b-c90acd7398cb\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/networkInterfaces/myNicVM1/ipConfigurations/ipconfig1",
  "loadBalancerBackendAddressPools": [
    {
      "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/loadBalancers/myLoadBalancer/backendAddressPools/myBackEndPool",
      "resourceGroup": "CreatePubLBQS-rg"
    }
  ],
  "name": "ipconfig1",
  "primary": true,
  "privateIpAddress": "10.1.0.4",
  "privateIpAddressVersion": "IPv4",
  "privateIPAllocationMethod": "Dynamic",
  "provisioningState": "Succeeded",
  "resourceGroup": "CreatePubLBQS-rg",
  "subnet": {
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/virtualNetworks/myVNet/subnets/myBackEndSubnet",
    "resourceGroup": "CreatePubLBQS-rg"
  },
  "type": "Microsoft.Network/networkInterfaces/ipConfigurations"
}
{
  "etag": "W/\"b4f6fddc-e20f-4493-8b1e-eba6ad57c59b\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/networkInterfaces/myNicVM2/ipConfigurations/ipconfig1",
  "loadBalancerBackendAddressPools": [
```

Create NAT gateway

```
az network public-ip create --resource-group CreatePubLBQS-rg --name myNATgatewayIP --sku Standard --zone 1 2 3
```

```
odl_user [ ~ ]$ az network public-ip create --resource-group CreatePubLBQS-rg --name myNATgatewayIP --sku Standard --zone 1 2 3
{
  "publicIp": {
    "ddosSettings": {
      "protectionMode": "VirtualNetworkInherited"
    },
    "etag": "W/\"8f93b00e-5ff8-4ab2-b175-e43bc72ac0f7\"",
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/publicIPAddresses/myNATgatewayIP",
    "idleTimeoutInMinutes": 4,
    "ipAddress": "40.88.223.67",
    "ipTags": [],
    "location": "eastus",
    "name": "myNATgatewayIP",
    "provisioningState": "Succeeded",
    "publicIPAddressVersion": "IPv4",
    "publicIPAllocationMethod": "Static",
    "resourceGroup": "CreatePubLBQS-rg",
    "resourceGuid": "e3f6fc98-5613-435b-b266-2bade6831c5f",
    "sku": {
      "name": "Standard",
      "tier": "Regional"
    },
    "type": "Microsoft.Network/publicIPAddresses",
    "zones": [
      "1",
      "2",
      "3"
    ]
  }
}
```

Create NAT gateway resource

```
az network nat gateway create --resource-group CreatePubLBQS-rg --name myNATgateway --public-ip-addresses myNATgatewayIP --idle-timeout 10
```

```
odl_user [ ~ ]$ az network nat gateway create --resource-group CreatePubLBQS-rg --name myNATgateway --public-ip-addresses myNATgatewayIP --idle-timeout 10
{
  "etag": "W/\"cf3280a8-a773-4bd2-8035-e01f2903f9a4\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/natGateways/myNATgateway",
  "idleTimeoutInMinutes": 10,
  "location": "eastus",
  "name": "myNATgateway",
  "provisioningState": "Succeeded",
  "publicIpAddresses": [
    {
      "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/publicIPAddresses/myNATgatewayIP",
      "resourceGroup": "CreatePubLBQS-rg"
    }
  ],
  "resourceGroup": "CreatePubLBQS-rg",
  "resourceGuid": "3f45aa48-f1f9-4d86-8739-463adfc4cfc3",
  "sku": {
    "name": "Standard"
  },
  "type": "Microsoft.Network/natGateways"
}
```

```
odl_user [ ~ ]$
```

Associate NAT gateway with subnet

```
az network vnet subnet update --resource-group CreatePubLBQS-rg --vnet-name myVNet --name myBackendSubnet --nat-gateway myNATgateway
```

```
odl_user [ ~ ]$ az network vnet subnet update --resource-group CreatePubLBQS-rg --vnet-name myVNet --name myBackendSubnet --nat-gateway myNATgateway
{
  "addressPrefix": "10.1.0.0/24",
  "delegations": [],
  "etag": "W/\"2369295e-fadc-491e-828b-845bfdd90502\"",
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/virtualNetworks/myVNet/subnets/myBackendSubnet",
  "ipConfigurations": [
    {
      "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CREATEPUBLBQS-RG/providers/Microsoft.Network/networkInterfaces/MYNICVM1/ipConfigurations/IPCONFIG1",
      "resourceGroup": "CREATEPUBLBQS-RG"
    },
    {
      "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CREATEPUBLBQS-RG/providers/Microsoft.Network/networkInterfaces/MYNICVM2/ipConfigurations/IPCONFIG1",
      "resourceGroup": "CREATEPUBLBQS-RG"
    }
  ],
  "name": "myBackendSubnet",
  "natGateway": {
    "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Network/natGateways/myNATgateway",
    "resourceGroup": "CreatePubLBQS-rg"
  },
  "privateEndpointNetworkPolicies": "Disabled",
  "privateLinkServiceNetworkPolicies": "Enabled",
  "provisioningState": "Succeeded",
  "resourceGroup": "CreatePubLBQS-rg",
  "type": "Microsoft.Network/virtualNetworks/subnets"
}
odl_user [ ~ ]$
```

Install IIS

```
array=(myVM1 myVM2)
for vm in "${array[@]}"
do
  az vm extension set --publisher Microsoft.Compute --version 1.8 --name CustomScriptExtension --vm-name $vm --resource-group CreatePubLBQS-rg --settings '{"commandToExecute":"powershell Add-WindowsFeature Web-Server; powershell Add-Content -Path \"C:\\inetpub\\wwwroot\\Default.htm\" -Value $($env:computername)}'
done
```

```

odl_user [ ~ ]$ array=(myVM1 myVM2)
for vm in "${array[@]}"
do
    az vm extension set --publisher Microsoft.Compute --version 1.8 --name CustomScriptExtension --vm-name $vm --resource-group CreatePubLBQS-rg --settings '{"commandToExecute":"powershell Add-WindowsFeature Web-Server; powershell Add-Content -Path \"C:\\inetpub\\wwwroot\\Default.htm\" -Value $($env:computername)}'
done
{
  "autoUpgradeMinorVersion": true,
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Compute/virtualMachines/myVM1/extensions/CustomScriptExtension",
  "location": "eastus",
  "name": "CustomScriptExtension",
  "provisioningState": "Succeeded",
  "publisher": "Microsoft.Compute",
  "resourceGroup": "CreatePubLBQS-rg",
  "settings": {
    "commandToExecute": "powershell Add-WindowsFeature Web-Server; powershell Add-Content -Path \"C:\\inetpub\\wwwroot\\Default.htm\" -Value $($env:computername)"
  },
  "type": "Microsoft.Compute/virtualMachines/extensions",
  "typeHandlerVersion": "1.8",
  "typePropertiesType": "CustomScriptExtension"
}
{
  "autoUpgradeMinorVersion": true,
  "id": "/subscriptions/baba3a9a-2ccf-483d-a1b9-8d09ae11d3c5/resourceGroups/CreatePubLBQS-rg/providers/Microsoft.Compute/virtualMachines/myVM2/extensions/CustomScriptExtension",
  "location": "eastus",
  "name": "CustomScriptExtension",
  "provisioningState": "Succeeded",
  "publisher": "Microsoft.Compute",
  "resourceGroup": "CreatePubLBQS-rg",
  "settings": {
    "commandToExecute": "powershell Add-WindowsFeature Web-Server; powershell Add-Content -Path \"C:\\inetpub\\wwwroot\\Default.htm\" -Value $($env:computername)"
  },
  "type": "Microsoft.Compute/virtualMachines/extensions",
  "typeHandlerVersion": "1.8",
}

```

Test the load balancer

get the public IP

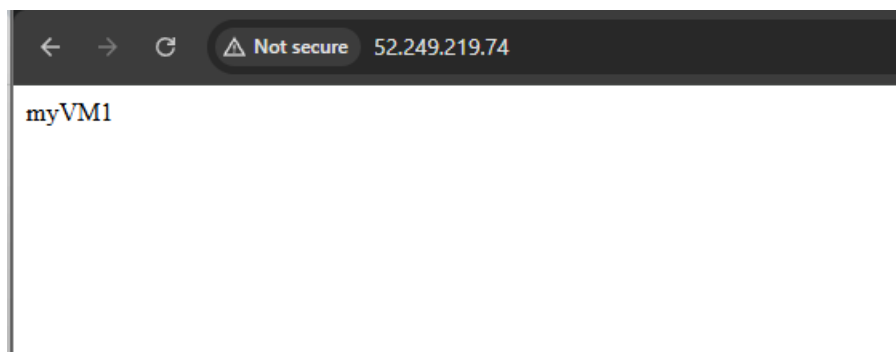
```
az network public-ip show --resource-group CreatePubLBQS-rg --name myPublicIP --query ipAddress --output tsv
```

```

odl_user [ ~ ]$ az network public-ip show --resource-group CreatePubLBQS-rg --name myPublicIP --query ipAddress --output tsv
52.249.219.74
odl_user [ ~ ]$

```

Browse using public ip



Extend the load balancer with backend pool and frontend IP

Define the Health probe